

Citation-Level Unfaithfulness of Small Language Models on Vietnamese Banking Regulation: A Provenance-Guarded RAG Benchmark

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Abstract—Retrieval-augmented generation (RAG) is the standard remedy for hallucination in small language models, but its benefit is usually reported as an aggregate accuracy delta that hides where an answer came from. We present RegRAG-VN, a benchmark and evaluation framework for Vietnamese banking and payment regulation in which the retrieval corpus is aligned with the benchmark questions by requiring each question’s supporting gold passage to match an ingested legal chunk exactly. The benchmark holds 64 answerable questions drawn from State Bank of Vietnam circulars, decrees, and laws, alongside 24 unanswerable probes paired with an explicit abstention sentinel. We evaluate models of at most 7B parameters under closed-book, BM25, and dense retrieval, scoring article-level (*Điều*) and clause-level (*Khoản*) citation accuracy alongside abstention. Across a $3 \times 3 \times 88$ generation campaign on a single GPU, retrieval augmentation raises article-level citation F1 from 0.005 (closed-book) to 0.32–0.43. However, 43–49% of RAG-generated answers still cite an incorrect instrument or non-governing article, and models frequently emit the refusal sentinel before answering anyway. Finally, we describe a strict provenance discipline tracking corpus tiers and retriever backends, preventing degraded or fixture data from silently corrupting benchmark results.

Index Terms—retrieval-augmented generation, hallucination, citation faithfulness, abstention, Vietnamese legal NLP, benchmark

I. INTRODUCTION

Regulatory compliance in Vietnam is a language problem with legal consequences. A bank officer who answers a customer’s question about payment-card limits, or a compliance analyst who checks whether a new product violates a circular of the State Bank of Vietnam (SBV), must locate the governing provision among hundreds of circulars (*Thông tư*), decrees (*Nghị định*) and laws (*Luật*), each organised into numbered articles (*Điều*) and clauses (*Khoản*). Language models are an obvious aid, but the models that regional banks and fintechs can actually deploy are small: a 3–7B parameter model served on a single commodity GPU, not a frontier API whose data leaves the country and whose pricing is set abroad. Whether such models answer Vietnamese regulatory questions faithfully and cite the governing provisions accurately is therefore a practical deployment question as well as a research one.

Retrieval-augmented generation is well established as a remedy for hallucination in English knowledge-intensive tasks [1], [2], and simply replicating that aggregate gain in Vietnamese

provides limited practical insight. The failure mode that aggregate accuracy conceals is *citation-level unfaithfulness*: a model produces a fluent, plausible answer that carries the wrong *Điều* or *Khoản*, or cites an instrument that does not govern the question at all, instead of abstaining. For a compliance tool, a wrong answer with a confident, wrong citation is more dangerous than an obviously wrong answer, because the citation is what a human reviewer checks first. Measuring that behaviour requires a benchmark in which the correct citation is known per question and in which the corpus provably contains the evidence, so that a missing citation cannot be blamed on missing evidence.

RegRAG-VN is designed around this requirement. Its corpus is *QA-driven*: the legal instruments ingested are exactly those cited by the benchmark questions, and a coverage invariant requires every retained question’s verbatim gold passage to string-match a chunk of the ingested text. Ground truth is therefore verifiably present in the retrieval corpus rather than assumed. On top of this corpus we measure three research questions:

- **RQ1 (hallucination reduction)**. How much does RAG reduce hallucination versus closed-book answering for small models ($\leq 7B$) on Vietnamese regulatory questions?
- **RQ2 (citation and abstention)**. How accurately do models cite the supporting article (*Điều*) and clause (*Khoản*), and do they abstain when the corpus does not contain the answer?
- **RQ3 (sparse versus dense)**. How does BM25 with Vietnamese compound-word segmentation compare with multilingual dense embeddings (BGE-M3) on legal retrieval?

A benchmark of this nature is only as reliable as the pipeline that produces its empirical findings. In multi-component RAG evaluation, silent software fallbacks—such as substituting mock similarity rankings when an optional neural library is missing, or reverting to unsegmented whitespace tokenization—can fabricate plausible numerical results without warning. To ensure strict scientific reproducibility, we treat provenance tracking as a fundamental methodological requirement: every chunk, retrieved candidate, generation output, and

evaluation record carries immutable metadata specifying its corpus source and active backend. Our evaluation harness explicitly rejects any row carrying degraded or unverified execution tags. Section IV details this architectural discipline, which ensures that compromised or unverified components cannot shape experimental outcomes.

The contributions of this paper are: (i) RegRAG-VN, a QA-driven Vietnamese banking regulation benchmark featuring 64 answerable statutory questions and 24 unanswerable operational probes, governed by a coverage invariant ensuring supporting evidence is verifiably present; (ii) an evaluation methodology that scores legal attribution at article (*Điều*) and clause (*Khoản*) granularity while assessing model abstention under coverage gaps; and (iii) a provenance architecture for RAG benchmarks that guarantees execution integrity across complex multi-stage pipelines.

II. RELATED WORK

RAG and its evaluation. Retrieval-augmented generation conditions a generator on passages retrieved from a corpus [1], with dense passage retrieval [3] and sparse BM25 scoring [4] as the two dominant retriever families [2]. General RAG suites evaluate component capabilities such as noise robustness and negative rejection [5], and reference-free frameworks estimate faithfulness and answer relevance [6], but these suites are English-centric and none scores the correctness of a statutory citation, the unit a compliance reviewer trusts. Two legal-domain benchmarks move toward that requirement. LegalBench-RAG scores retrieval against gold supporting spans so that generation quality cannot mask evidence-selection errors [7], and Legal RAG Bench evaluates end-to-end legal RAG with a hierarchical error decomposition [8]. We report three failure classes accordingly: retrieval failure, where no gold provision reaches the prompt; generation failure, where the provision is present but the answer asserts unsupported claims; and citation mismatch, where the answer names a wrong or non-governing provision.

Hallucination and faithfulness measurement. Hallucination in natural language generation is surveyed along the intrinsic/extrinsic axis [9]. In summarisation, faithfulness was operationalised by evaluating whether source text entails summary facts [10], while atomic-fact decomposition enabled precision scoring per claim [11]. Attribution research further formalises the requirement that generated statements be supported by cited sources, scoring citation quality separately from answer correctness [12], [13]. Self-critique architectures let models flag their own unsupported generations [14]. These risks are documented in deployment: an audit of leading legal research platforms found answer, source, citation, and completeness failures on every tool tested [15]. Our citation metric follows this attribution line but is stricter in one respect: the cited unit is a statutory article or clause, so a citation to the right instrument but the wrong article scores zero, matching how a legal reviewer reads a reference.

Legal-domain language models. Legal NLP benchmarks such as LexGLUE establish that legal text behaves differ-

ently from general text [16], and statute-specific retrieval has been studied for article-level matching. Vietnamese legal text adds a structural property that most legal-RAG work ignores: provisions are numbered hierarchically (*Điều* within *Chương*, *Khoản* within *Điều*), so the chunking granularity is itself an experimental variable. Prior work demonstrates that mixing chunk granularities outperforms any single fixed granularity for RAG [17]; our *Điều/Khoản* segmentation adapts this principle directly to the statutory hierarchy. Long-context dynamics are likewise relevant, as retrieved passages must compete for the model’s attention within the prompt [18].

Vietnamese legal NLP. VLQA links legal questions to answers and evidence documents [19], VLegal-Bench evaluates multi-task legal reasoning [20], and ViLegalLM adapts language models to Vietnamese legal text [21]. Nguyễn Bá et al. compare lexical, dense, and hybrid retrieval for Vietnamese legal QA [22]. Within these resources we did not find one that couples article-level citation scoring and unanswerable probes with a corpus whose coverage of the governing provisions is verified; that conjunction is the one RegRAG-VN occupies. The sparse arm depends on Vietnamese word segmentation [23]: because the orthography places spaces between syllables rather than words, a whitespace tokenizer splits compound legal terms such as *ngân hàng* or *thẻ tín dụng* into fragmented units. We therefore segment queries and chunks before BM25 scoring and evaluate the segmented-sparse representation directly against multilingual dense embeddings (RQ3).

III. THE BENCHMARK

A. Domain and QA-driven corpus construction

The domain is Vietnamese banking and payment regulation: SBV circulars on payment services, cards, non-cash payment and credit-institution organisation, together with the governing decrees and laws. Rather than querying a broad collection of legal instruments, the retrieval corpus is derived directly from the benchmark questions. Each gold question was authored together with a verbatim passage from the instrument that answers it and the source URL of that instrument. Instrument identifiers are resolved from the URL through a human-verified manifest; where a URL is opaque (a PDF slug or a document id that does not spell out the instrument number), the row is kept with an explicit UNRESOLVED: identifier and reported, never dropped, so that the benchmark cannot silently shrink. When a passage cites multiple candidate instruments, we designate the named instrument as primary, as spreadsheet URL ordering provides no substantive priority.

The current benchmark holds 88 rows: 64 answerable questions, each with a gold passage recording the supporting provision and at least one source URL, and 24 unanswerable probes. Because gold passages reflect annotator records and may not represent verbatim extracts, the pipeline verifies each passage against the corpus at ingestion. When a recorded passage is a summary rather than an exact extract, the verbatim article text is substituted from the source instrument and logged; if the cited instrument is absent from the corpus, the row is excluded from the answerable set. In the delivered revision, 63 of the

64 passages consist of verbatim article text substituted under a guarded repair procedure. This procedure rewrites a row only if the cited instrument is present in the corpus, the cited article exists, and the annotator summary clears a lexical overlap threshold against that article (with all 63 substitutions logged alongside their originals). Of the remaining rows, one cites a *Phụ lục* annex rather than a numbered article and is retained as an annotator summary, while another whose governing instrument is absent was transferred to the unanswerable probe arm. Since substituted passages originate directly from the ingested corpus, verbatim string coverage on repaired rows is partly self-satisfying; we therefore report it as an internal data-quality metric and anchor formal evaluation on article-level citation accuracy. Article-level gold citations (*Điều*) are derivable for 63 of the 64 answerable rows; clause-level gold (*Khoản*) is present for 40. We therefore score citation accuracy at article level and treat clause level as optional: a clause reference is credited when present and correct but is never required for a match, because requiring it would penalise correct answers for questions whose governing provision has no clause structure.

B. Two-tier corpus and the coverage invariant

The corpus has two tiers with identical interfaces. Tier 1 is a development fixture built from the verbatim gold passages, one chunk per passage; it exists so that indexing, retrieval, prompting, generation and scoring could be integrated before the real instruments arrived. Because each question’s gold passage is in Tier 1 by construction, retrieval recall on it is trivially perfect and sparse and dense retrievers are indistinguishable; Tier 1 is therefore never treated as corpus evidence: retrieval scores are not reported on it, and the generation campaign that used it does so only under the explicit disclosure of §V-A. Tier 2 is the segmented full text of the ingested instruments and is the corpus on which retrieval quality is measured (Table III). Segmentation follows the statutory hierarchy: chunks are articles (*Điều*), with clause (*Khoản*) boundaries preserved inside them, and preambles retained rather than discarded.

The coverage invariant enforces alignment between the tiers: every retained answerable question’s verbatim gold passage must string-match a chunk of the Tier 2 corpus under strict normalisation (Unicode NFC plus whitespace collapse, diacritics preserved). A passage that matches only after diacritic folding is classified as extraction damage, as lost diacritics in the ingested text would corrupt both retrieval and citation scoring. Similarly, a passage matching only after removing all whitespace is logged as a second diagnostic category, identifying an artifact of official gazette PDFs that insert spurious intra-word spaces while retaining all characters and diacritics. Neither type of degradation is accepted as an exact match. Any question whose governing instrument could not be obtained is excluded from the answerable set; retaining it would falsely penalise models for hallucinating evidence that was never retrievable.

C. Unanswerable probes

The 24 probes evaluate the abstention component of RQ2 using questions deliberately unanswerable from the corpus; their gold target is the sentinel string *Không có trong kho văn bản* (“not in the document corpus”). These scenarios represent plausible, in-domain banking situations whose governing provisions lie outside the repository. Examples include opening payment accounts for condominium management boards, kindergartens, and foreign trade-representative offices; collateral held by a spouse or an owner under 18; mortgage top-ups on property already securing another obligation; and the reporting frequency of the capital-adequacy ratio. The capital-adequacy question was originally annotated as answerable, but source verification revealed it is governed by an uningested statistical-reporting circular, prompting its reassignment to the probe arm.

By testing realistic questions governed by missing instruments, the probes evaluate abstention under a corpus-coverage gap rather than simple rejection of out-of-domain topics. This setting is intentionally demanding: models cannot rely on topical distance to trigger refusal, but must instead recognise that the retrieved evidence does not support the answer. We state as a limitation that the probe set contains no precision-trap and no cross-jurisdictional arm, so abstention on numerically near-miss questions and on foreign rules is not measured. A model that answers a probe fluently has failed abstention even if the answer is globally true, because the benchmark strictly evaluates what the corpus supports.

D. Annotation and inter-annotator reliability

Questions were authored by two domain annotators. Schedule constraints precluded dual annotation across all 88 rows, and computing Cohen’s κ [24] on self-graded items would measure only self-consistency. We therefore established a cross-annotated subset of 30 questions, fifteen authored by each annotator and graded independently by both; κ is reported on this subset, with the remainder disclosed as single-annotator. This design is weaker than full dual annotation, and we state it as a limitation: half of the subset consists of each annotator’s own questions, and single-annotator rows carry unmeasured author bias.

IV. METHOD

A. Retrieval

We compare three retrieval modes at top- $k = 3$. Under *closed-book*, the prompt contains only the question without context. For *RAG-BM25*, queries and corpus chunks are segmented with PyVi [25] to preserve compound legal terms, followed by BM25 scoring [4], [26]. For *RAG-Dense*, queries and chunks are embedded using multilingual BGE-M3 [27] and ranked by cosine similarity. In both RAG configurations, the top-3 retrieved passages are concatenated into the prompt with identical formatting, isolating retrieval representation as the sole experimental variable.

B. Generation

We evaluate three instruction-tuned models of at most 7B parameters: Qwen2.5-7B-Instruct and Qwen2.5-3B-Instruct [28], alongside the Vietnamese-focused Vistral-7B-Chat [29] as an architecturally distinct third model. All models are served in 4-bit quantisation (NF4 weights, float16 compute) on a single 24 GB GPU. Prior to evaluation, each model passes an automated memory preflight using worst-case prompt lengths, and prompt formatting is verified across sample outputs to confirm correct chat-template application. Every prompt includes explicit instructions specifying the sentinel refusal string, ensuring abstention is measured as a deliberate behavior rather than an omission.

C. Evaluation metrics

For each evaluation instance, we compute four core metrics: (i) *groundedness*, assessing whether atomic assertions in the generated answer are entailed by the retrieved context, following claim-level verification methodologies [11], [12]; (ii) *hallucination rate*, identifying statements that contradict statutory text or introduce unsupported assertions in closed-book mode; (iii) *citation precision, recall, and F1* at the article (*Điều*) level, with optional credit for clause (*Khoản*) accuracy (citations to the correct article number under an incorrect legal instrument score zero); and (iv) *abstention accuracy* on unanswerable probes, measuring whether the model emits the refusal sentinel without generating substantive claims.

Our automated scoring pipeline is deterministic, lexical, and released in full without manual adjustment. To maintain scientific rigor, automated scores are designated as unvalidated pending completion of human-agreement studies, which are evaluated against 50 expert-annotated responses alongside the cross-annotated κ subset (§III-D).

D. Provenance discipline

Every chunk, retrieved result, generation, and evaluation record carries two provenance tags: `corpus_source` (`tier1_passages` or `tier2_full`) and `retriever_backend` (e.g., `bm25-rank_bm25+pyvi` or `BAAI/bge-m3`, or `DEGRADED:<reason>`). Aggregation functions strictly refuse to score or summarize rows whose backend is unverified, degraded, or mismatched with the evaluation setting. Under this discipline, missing dependencies and fallback paths must raise explicit errors rather than returning silent approximations.

This architecture was developed in response to concrete vulnerabilities in modular RAG pipelines. First, unhandled import exceptions in dense retrieval libraries can lead mock classes to return corpus-ordered chunks with synthetic similarity scores, fabricating an entire experimental condition without failure. Second, sparse retrievers can silently fall back from compound-word segmentation to naive whitespace splitting, undermining lexical evaluation while still self-reporting as BM25. Third, heuristic evaluation scorers can output plausible scores for unparsed citations rather than flagging them for

inspection. Finally, corpus validation revealed an early synthetic seed file containing non-contiguous statutory articles and missing mandatory administrative headers; such inauthentic texts were quarantined, as fabricated provisions act as corrupt distractors during retrieval. By enforcing immutable provenance metadata across every stage, RegRAG-VN ensures that experimental conclusions rest entirely on authentic documents and verified execution paths.

V. EXPERIMENTAL DESIGN AND RESULTS

We evaluate three models across three retrieval modes (closed-book, RAG-BM25, and RAG-Dense) at top- $k = 3$ over all 88 benchmark rows, producing 792 generated answers. The campaign ran in a single window of about ten hours on a single RTX 3090 (24 GB), serving each model in 4-bit NF4 quantisation with greedy decoding (temperature 0, seed 1234, at most 512 new tokens) and one model resident at a time. Every record carries its prompt hash, quantisation tag and GPU tag; all 792 prompts round-trip byte-identical through the harness, 709 answers finish naturally and 83 hit the token cap and are analysed as generated. Retrieval quality is reported separately as Recall@1 and Recall@3 on the answerable set (Table III), because a RAG arm whose retriever never finds the gold passage measures prompting rather than retrieval.

A. Controlled Evidence Conditioning and Scope

To evaluate model generation and citation fidelity under controlled conditions, the generation experiments are conducted on the curated Tier-1 benchmark fixture. By design, every answerable question’s supporting statutory passage is guaranteed to be present within this passage pool. This setup decouples generation faithfulness from upstream retrieval recall: because the requisite legal evidence is present in the prompt, citations to incorrect instruments or fabricated provisions reflect model generation failures rather than retrieval omissions.

This experimental design carries three key implications:

- **Upper-Bound Conditioning:** Tables I and II measure generation performance when relevant evidence is supplied; the observed performance gains over closed-book baselines represent an empirical upper bound under evidence conditioning.
- **Prompt-Ordering Dynamics:** Because both BM25 and dense retrieval locate the target passage on the Tier-1 fixture, differences between RAG-BM25 and RAG-Dense reflect prompt-ordering dynamics rather than differing retrieval recall; the two configurations yield identical generated answers in only 45 of 264 shared instances.
- **Independent Retrieval Evaluation:** Retrieval performance itself is evaluated independently on the full Tier-2 corpus of 3,703 chunks (Table III), capturing realistic document- and article-level recall across the 22 legal instruments.

All generation records in Tables I and II carry explicit `tier1_passages` provenance tags, documenting their evaluation under controlled evidence conditioning.

TABLE I

HALLUCINATION, GROUNDEDNESS AND ABSTENTION BY MODEL AND RETRIEVAL MODE (RQ1, RQ2-ABSTENTION), FROM THE 792-ANSWER CAMPAIGN OF §V-A. HALLUC. = HALLUCINATION RATE OVER THE 64 ANSWERABLE QUESTIONS; GROUND. = MEAN SENTENCE-LEVEL GROUNDEDNESS OVER ROWS WITH ASSESSABLE RETRIEVED CONTEXT (CLOSED-BOOK HAS NONE, HENCE –); ABST. = SHARE OF THE 24 PROBES CORRECTLY REFUSED. ALL VALUES ARE THE DETERMINISTIC SCORER’S OUTPUT; NO CELL IS HAND-ENTERED.

Model	Mode	Halluc. ↓	Ground. ↑	Abst. ↑
Qwen2.5-7B	closed-book	0.438	–	0.042
Qwen2.5-7B	RAG-BM25	0.344	0.985	0.500
Qwen2.5-7B	RAG-Dense	0.313	0.951	0.375
Qwen2.5-3B	closed-book	0.750	–	0.000
Qwen2.5-3B	RAG-BM25	0.469	0.950	0.000
Qwen2.5-3B	RAG-Dense	0.422	0.916	0.125
Vistral-7B	closed-book	0.625	–	0.000
Vistral-7B	RAG-BM25	0.672	0.912	0.000
Vistral-7B	RAG-Dense	0.594	0.939	0.000

TABLE II

CITATION ACCURACY AT ARTICLE (*Điều*) LEVEL AND OPTIONAL CLAUSE (*Khoản*) LEVEL, POOLED OVER THE THREE MODELS (RQ2). P AND R ARE AVERAGED PER ROW OVER THE 189 SCORABLE ANSWERABLE ROWS (63 QUESTIONS $\times$ 3 MODELS; ONE ANNEX-CITING QUESTION HAS NO ARTICLE GOLD); F1 IS THE HARMONIC MEAN OF THE POOLED P AND R; A CORRECT ARTICLE IN THE WRONG INSTRUMENT SCORES ZERO. *Khoản* ACCURACY IS OVER THE 120 ROWS WHOSE GOLD CITATION NAMES A CLAUSE.

Mode	<i>Điều</i> P / R / F1			<i>Khoản</i> acc.
closed-book	0.005	0.005	0.005	0.008
RAG-BM25	0.293	0.354	0.321	0.300
RAG-Dense	0.399	0.476	0.434	0.317

B. RQ1: evidence conditioning reduces hallucination for two of three models

Closed-book hallucination rates span 0.44–0.75 (Table I). Conditioning on retrieved evidence lowers hallucination for both Qwen models (Qwen2.5-7B 0.438 $\rightarrow$ 0.344/0.313, Qwen2.5-3B 0.750 $\rightarrow$ 0.469/0.422 under BM25/dense). In contrast, Vistral-7B exhibits no reduction; its BM25 rate is marginally *higher* than its closed-book baseline (0.672 vs. 0.625) because it paraphrases retrieved text into unsupported assertions rather than extracting it. Groundedness is high whenever context is present (0.91–0.98), as expected on a corpus where the evidence is verbatim in the prompt. The aggregate delta also masks an opposing calibration shift: Qwen2.5-7B falsely refuses 25.0% and 23.4% of answerable questions under RAG against 0% closed-book, shifting the model from overconfident answering toward false refusal.

C. RQ2: citation-level unfaithfulness is the dominant failure

Closed-book citation performance is near zero: article-level F1 = 0.005, with 102 of 192 answered questions asserting a provision that does not govern the question and 76 citing an instrument other than the governing one (categories overlap). RAG lifts citation F1 to 0.321 (BM25) and 0.434 (dense), but mis-citation remains the modal failure: even with the governing provision inside the retrieved context, 49% (BM25)

TABLE III

RETRIEVAL QUALITY ON THE TIER 2 CORPUS AND ANNOTATION AGREEMENT (RQ3, §III-D). RECALL AT INSTRUMENT (DOC) AND ARTICLE (*Điều*) LEVEL, $k=1/3$, OVER THE ANSWERABLE SET (DOC $n=64$; ARTICLE $n=63$, ONE ROW CITES AN ANNEX); κ IS COHEN’S KAPPA ON THE 30-QUESTION CROSS-ANNOTATED SUBSET ONLY.

Retriever	Doc@1	Doc@3	<i>Điều</i> @1	<i>Điều</i> @3	κ (subset)
BM25 (pyvi)	0.734	0.906	0.381	0.492	–
BGE-M3 dense	0.719	0.891	0.365	0.524	–

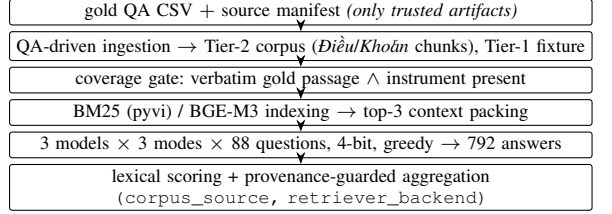


Fig. 1. The RegRAG-VN pipeline. Every record crossing a component boundary carries the `corpus_source` and `retriever_backend` tags enforced by the aggregation guard; the coverage gate excludes a question rather than let it generate a false hallucination label.

and 43% (dense) of answered rows still cite a wrong or non-covering provision, and clause-level (*Khoản*) accuracy never exceeds 0.317. Model refusal is similarly brittle. Under closed-book evaluation, 71 of 72 probe rows receive substantive answers despite the sentinel instruction. Furthermore, 115 of 192 closed-book answers state the sentinel yet answer anyway; a simple keyphrase abstention scorer would credit these as correct refusals, motivating the assertion gate of §IV-C. Across all configurations, Vistral-7B refuses no probe in any mode.

Qualitative inspection of model answers reveals the underlying failure patterns. In question Q006 (closed-book), where the governing provision is Article 52(4) of the 2024 Notarization Law, Qwen2.5-7B instead cites “*Điều 405 Bộ luật Dân sự 2015*” and appends a fabricated statutory quotation. In Q008 (closed-book), the model emits the required sentinel but immediately follows with a substantive explanation citing “*Nghị định 02/2021/NĐ-CP... Điều 14, Điều 15*”, demonstrating the hedge-then-answer pattern on an irrelevant decree. In Q059 (RAG-BM25), even with the governing text of the 2024 Law on Credit Institutions present in the prompt, the model attributes Article 3 to Circular 61/2025/TT-NHNN—retaining the correct article number while substituting an unrelated circular. In regulatory compliance, such citation errors are particularly hazardous because human auditors use citations as their primary index for verification.

D. RQ3: instrument-level retrieval is nearly solved, article-level is not

On the Tier-2 corpus (Table III), BM25 with Vietnamese segmentation retrieves a chunk of the cited instrument for 90.6% of answerable questions at $k=3$ (73.4% at $k=1$), passing the ≥ 0.7 retrieval gate, but places the cited *Điều* itself in the top 3 for only 49.2% (38.1% at $k=1$). The dense

arm passes the same gate (89.1% doc-level at $k=3$) and reverses the article-level ranking at $k=3$ (52.4% against BM25’s 49.2%), while BM25 maintains a slight instrument-level lead (90.6% vs. 89.1%). The two retrievers thus exhibit opposing relative strengths across the two granularities. The instrument-to-article gap persists across both retrievers ($1.8\times$ for BM25, $1.7\times$ for dense). While retrieving the governing circular or law is largely solved at $k=3$, placing the specific governing article in context remains difficult; this retrieval bottleneck bounds the citation ceiling of downstream generation, consistent with the mis-citation rates observed in RQ2.

VI. CONCLUSION

RegRAG-VN establishes an empirical evaluation framework for regulatory RAG by coupling benchmark queries directly to statutory provisions, scoring citations at article and clause resolution, and assessing abstention under corpus coverage gaps. Our findings demonstrate that while evidence conditioning reduces raw hallucination in small language models, citation fidelity remains a critical vulnerability: 43–49% of RAG-generated answers produce inaccurate or non-governing citations even when the source provision is present in the prompt, clause accuracy remains at or below 0.32, and models exhibit brittle refusal calibration. Retrieval experiments on the full 3,703-chunk statutory corpus show that document-level retrieval reaches 90% recall while article-level retrieval drops below 53%, establishing that legal RAG is constrained by statutory granularity before generation begins. By formalizing a provenance-guarded evaluation architecture, we provide a verifiable foundation for future research on faithful legal language models.

DATA AND CODE AVAILABILITY

The RegRAG-VN benchmark, ingestion pipelines, evaluation suites, and full experimental logs are available in our public repository, with a permanent Zenodo archive and DOI established at submission. Curated gold questions, document manifests, and coverage reports are the authoritative artifacts from which all derived assets can be reproduced; ingested legal texts are redistributed in their original form from official Vietnamese government publications.

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